



FROM VISIBLE SPEECH TO VOICEPRINTS – THE MISSING LINK

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Abstract: This contribution is a sequel to this author’s presentation at HSCR 17 in Helsinki. At that point, a key report on voiceprint analysis from the time of WWII had not been accessible to the present author. Since then, access has been gained to this paper as well as other as yet unpublished sources. Both document different aspects of military use of the sound spectrograph and yet with somewhat unexpected results. Firstly, there is evidence that the sound spectrograph was not exclusively developed with voice comparison in mind. It was also alleged to serve as an important element in the decryption of German radio messages. Secondly, the early notion of “voiceprints” as outlined in the report in question is vastly different from Lawrence Kersta’s idea of their forensic use, which have so far been assumed to be based on the wartime work.

1 Introduction

1.1 Recapitulation and general setting

This paper is intended as a sequel to the one published two years ago at the workshop on the History of Speech Communication Research (HSCR) in Helsinki, which was entitled “From *visible speech* to voiceprints. Blessing and curse of sound spectrography” [2]. What was known of the development of the spectrograph was that engineers from Bell Telephone Laboratories had been working on a number of projects which were financed by the National Defense Research Committee (NDRC) of the Office of Scientific Research and Development at the time of WW II. The person responsible on the part of Bell Labs was C.H.G. Gray, who served as “Transmission Standards Engineer” [6]. The NDRC was represented by Dr. O.E. Buckley, Chairman of the Speech Secrecy section. [cf. 7, p. 1].¹

The projects covered in that paper were war projects and rated either *confidential* or *secret*. They were not declassified until 1960. From the documents publicly available, it is quite clear that “Dr. R. K. Potter invented the sound spectrograph” [7, p. 1]. The job was finished by the end of 1941 [7, p. 7]. More precisely, “[e]arly in 1941, a rough laboratory model of the sound spectrograph became available at Bell Telephone Laboratories Inc.” [7, p. 1]. The patent was not filed until April 14, 1942, though; it was issued on July 16, 1946 [4, p. 293]. A secret wartime document linking the development of the spectrograph to speaker identification was cited by [25], but this author had been unable to retrieve the document by the time of the previous publication. The description of the rationale behind the construction of the sound spectrograph which was contained in that citation differed considerably from the official version [21, 22, 23]. The latter insisted that sound spectrography had been developed in order

¹ Even today it is not entirely uncommon for phonetic research to be financed by grants from the military. Examples are the research on speech under stress which was to shed light on the stress level of fighter pilots based on their vocal behavior or the research on dolphin communication, which was important in conjunction with using dolphins as messengers in a war [cf. e.g. 10, 11].

to allow congenitally deaf people to be able to use the telephone, and Potter, Kopp, and Kopp Green went through great pains to even publish papers and a bogus monograph on this topic shortly after WW II [21, 22]. Tosi, on the other hand, cited a report by Gray and Kopp of 1944 which linked the development of the spectrograph to speaker identification [2].

1.2 The mysterious report

Shortly after the 2017 conference, Richard Sproat pointed out to this author that a copy of the mysterious report might be contained in the papers of the late Harriet Kopp-Green which had been archived at the Smithsonian Institute. This suggestion proved to be true, and she managed to secure a copy of the formerly secret report [9] as well as a total of three preliminary versions which had been drafted since Nov. 26, 1943 [8]. The first two drafts were entitled “Voiceprint Method of Identifying Individuals”; the third draft (dated June 10, 1944) and the final report were entitled “Voice Print Identification”. The different versions reflect the development of the methodology within the group of scientists involved and will be discussed elsewhere.

The final report entitled “Voice Print Identification” is dated August 18, 1944 and consists of 14 pages plus 16 pages of attachments [9].² It is signed by C.H.G. Gray and G.A. Kopp. The central claim made in the foreword [9, p. 0] is that “a speaker may be identified by his voice print.” However, this statement does not come without qualification: The authors point out that the report represents an interim stage of research and that a lot more work would have to be done before implementing it:

It should be appreciated that much work remains to be done to supplement the initial findings. [...] This report concludes the preliminary investigation of the possibilities of voiceprint identification. It has been prepared primarily for the information of military personnel. Further work in Bell Telephone Laboratories is contingent upon the military interest and this report is being submitted to enable a decision to be reached regarding the continuation or extension of this research. [9, Foreword, p. 0]

Gray and Kopp consider aural voice comparison a “conceivable” procedure, but describe it as “cumbersome and difficult” [9, p.1] and requiring a lot of training. They then proceed to point out the advantages of the spectrograph over an analysis by ear, but at the same time address some of the problems in speaker identification which forensic phoneticians are still struggling with today: adverse recording conditions, disguise, and within-speaker variability:

To determine similarities in speech over radio circuits subject to *different transmission conditions* and with talkers using *different microphones* under *different talking conditions* would require a highly trained ear.

Much more simple and quantitative than the aural scheme is a technique based upon the Sound Spectrograph, an instrument which makes permanent patterns of speech which can be examined visually in the way fingerprints and handwriting can be examined and can be filed as reference samples. [9, p. 1; emphasis mine, AB]

This passage is probably the origin of the fingerprint analogy. It is very clear from the rest of the report, though, that the idea of these researchers was not to create an analogy to fingerprinting in the sense that spectrographic images were to be compared solely by holistic visual inspection, but that they were proposing a parameter-driven approach. Neither is there any indication that the same ten words (viz. ten fingers) would suffice to draw conclusions on speaker identity, both those claims having been made by Lawrence Kersta (cf. 1.3 below). Rather, the spectrograms were to be used to build a database of enemy speakers against which any new recording was matched.

² Only Tables I to III (= six pages total) were available to this author, though.

Gray and Kopp begin by addressing the issues of distortion and of disguise and proceed to develop an exhaustive list of parameters which would be robust to both. Then they discuss the key issue of any speaker identification, i.e. the individuality of the voice:

The individuality of a particular voice is determined by the factors effecting or controlling the speech process. These factors in general are coordinated in automatic speech habits. However, they are modified from time to time by mental and emotional states of being [9, p. 4].

This passage is highly important because it shows an understanding of “voice” as a habitual and thus variable rather than an anatomical and thus invariable concept. This understanding precludes what was later to become the voiceprint approach and corresponds to the perspective outlined by Francis Nolan [20] of speech as behavior with anatomy defining the limits of variation.

The systematic coverage of possible features in speaker identification by Grey and Kopp [9] distinguishes between variation within the voice “not characterized by any abnormality” (p.6) and “deviations in the speech processes” (p. 4). The description of the latter is organized along the lines of the speech production process and covers *respiration*, *phonation*, *resonation*, and *articulation*. All those concepts are briefly explained, giving non-experts a chance to understand the principles. Secondly, the authors describe how the features in question are represented in a spectrogram. Table 1, reproduced from the original report, lists the features which the authors consider to be accessible through visual inspection of spectrograms.

They go on to address the issue of intraspeaker variability (“normal day-to-day variations”; [9], p.8) and of speaker identification in a foreign language. What follows is a step-by-step description of how and by whom speaker identification work is to be carried out. They even propose building a database in order to assess the frequency of occurrence of the various speech patterns.

While the characteristics of “deviant” voices are easily recognizable in a spectrogram, “normal voices” present more of a problem:

There, of course, will be many individuals whose speech is either normal or so nearly normal that it is not characterized by any abnormality that can be used as a dependable basis for identification purposes. To distinguish between these individuals, as well as among those in a group possessing some particular characteristic speech deviation, it is necessary to examine voice prints in greater detail, making measurements of the various characteristics [...] [9, p. 6].

A detailed description of how to identify and measure these features is included. Table 2, which is taken from the final report, lists the features to be measured in the voice identification process. The list is very comprehensive and shows an excellent understanding of where to look for individuality in speech. In fact, it does not differ substantially from what was later to be known as the auditory-acoustic method of speaker identification [5, 18, see below]. The Gray and Kopp paper even goes beyond the early stages of the auditory-acoustic approach in seeking to establish a database which would allow for probability ratings based on the frequency of occurrence:

With further study of the above factors, it is conceivable that some sort of a probability table could be constructed which could be used as a guide to determine rather exactly the probability that the unknown was the same as that of some reference speaker (p.12).

Table 1. Elements in the speech process which were considered to be contained in spectrograms according to Gray and Kopp [9].

TABLE I

VOICE PRINT IDENTIFICATION CHART

Items in Speech Process

Recognizable in Voice Prints by Visual Examination

A. Respiration

1. Normal
2. Breathiness
3. Wavering
4. Spasmodic

B. Phonation

1. Normal
2. Monopitch
3. Extreme Variation in Pitch
4. Unusually High Pitch
5. Unusually Low Pitch
6. Voicing of Unvoiced Sounds
7. Unvoicing of Voiced Sounds
8. Glottal Stops

C. Resonation

1. Normal
2. Nasality
3. Negative Nasality
4. Falsetto
5. Muffled
6. Dissonant

D. Articulation

1. Normal
2. Sound Substitution
3. Sound Omission
4. Sound Addition
5. Sound Repetition
6. Block
7. Accent

Table 2. Parameters to be measured and compared in voice comparisons according to Gray and Kopp [9].

TABLE II	
VOICE PRINT IDENTIFICATION CHART	
Items in Speech Process	
Requiring Measurements or Direct Comparison of Voice Prints	
A. Entire Speech Sample	
1. Mean Fundamental Frequency or Pitch	
2. Range of Fundamental Frequency	
3. Melody	
4. Rhythm	
5. Rate	
6. Pause Length	
7. Relative Duration -- Vowel/Consonant	
8. Relative Amplitude -- Vowel/Consonant	
B. Vowels, Semi-Vowels or Voiced Consonants	
1. Number of Harmonics	
2. Missing Harmonics	
3. Highest Harmonic	
4. Frequency Pattern	
5. Duration	
6. Amplitude	
7. Unmarked Areas	
C. Consonants	
1. Frequency Position of Principal Zones of Energy	
2. Frequency Pattern	
3. Duration	
4. Amplitude	
5. Proximity to Vowel	
6. Unmarked Areas	

This list of acoustic parameters bears a striking similarity to the standard auditory-acoustic approach to forensic speaker identification which was established by Hermann Künzel at the forensic science laboratory of the German *Bundeskriminalamt* (Federal Criminal Police Office) in the 1980s [18]. It therefore seems worthwhile to compare the set of features to be included in voice comparison reports in the Gray and Kopp paper [9] to that published by Künzel [18] as well as to that which is still used by forensic practitioners today [5]. The former is the first monograph which outlines the auditory-acoustic method in detail; the latter is a compilation based on a survey in which 36 experts from 13 countries took part. The results are summarized in Table 3.

Table 3. Comparison of parameters used in speaker identification as listed in Gray & Kopp [9], Künzel [18], and Gold and French [5].

Element of speech production	Gray & Kopp (1944) deviations from normal = individual	Gray & Kopp (1944): variation within the normal voice	Künzel (1987)	Gold & French (2011)
respiration	Breathy		✓	✓
	Wavering		✓	✓
	Spasmodic		✓	✓
phonation	Monopitch	Mean F0	✓	✓
	Extreme variation in pitch	Range of F0	✓	✓
	Unusually high pitch	Melody	✓	✓
	Unusually low pitch	Rhythm	✓	✓
	Voicing of unvoiced sounds		✓	✓
	Unvoicing of voiced sounds		✓	✓
	Glottal stops		✓	✓
Resonation	Nasality		✓	✓
	Negative nasality		✓	✓
	Falsetto		✓	✓
	Muffled		✓	✓
	Dissonant		✓	✓
Articulation	Sound substitution		✓	✓
	Sound omission		✓	✓
	Sound addition		✓	✓
	Sound repetition		✓	✓
	Block		✓	✓
	Foreign Accent		✓	✓
	Regional Accent		✓	✓
Measurements: vowels		Number of harmonics		✓
		Missing harmonics		
		Highest harmonic		
		Frequency pattern		✓

		Duration		✓
		Amplitude		✓
		Unmarked areas		
Measurements: consonants		Frequency position of principal zones of energy		✓
		Frequency pattern		
		Duration		
		Amplitude		
		Proximity to vowel		
		Unmarked areas		
Measurements: entire sample ³		Speech rate	✓	✓
		Pause length		✓
		Relative duration vowel-consonant		
		Relative amplitude vowel-consonant		

This immediately raises the question of whether Künzel knew the Gray and Kopp report [9] when writing his book. While there is no proof one way or the other, the present author was working at the BKA forensic laboratory in the final stages of Künzel's book and never saw a copy of the report. What is to be noted is that in the early days of sound spectrography, the approach to speaker identification was very "modern", and the term "voice print" seems to have been a catchier way of selling the project (and the product) to the customers within the military rather than representing the scientists' conviction of an analogy between fingerprints and spectrograms.

The characteristics of any new spectrogram were to be compared with those on file, and the best match would be considered as originating from the same speaker. In view of the fact that the so-called *voiceprinters* later argued that actually listening to the tapes was not necessary, it is important to note that Gray and Kopp expressly refer to listening to the recordings as a final check:

[...] an auditory check on the identification can be obtained by playing the phonograph recordings of the unknown and of the person selected as most closely resembling the unknown, and listening for similarities and differences in these two recordings [9, p. 12].

One item which really seems unrealistic from today's point of view is the estimate of the amount of training required to become a proficient voice analyst.

It would seem likely that an individual with six weeks intensive training could, with the part time assistance of other intercept personnel in obtaining the phonograph records, carry on the identification work at a satisfactory rate under most field conditions. Some knowledge of the enemy language would be helpful but not necessary [9, p. 13].

³ One has to take into account that "entire sample" at that time was invariably much shorter than it is today since it took several minutes to produce a spectrogram of 2.4 seconds of speech.

This can be taken to cater to the expectations of the customers in the military. It would have probably not been possible to sell the product if the duration of the training phase had been realistically calculated.

In the Gray and Kopp report there is no mention of a need to validate their claims about the feasibility of voice identification. This may also be attributable to the time pressure during the final stages of WW II. The report concludes with a word of caution, though:

It is necessary to do considerably more work to establish the relative importance which should be assigned to the several identifying characteristics so that the identification process might be simplified and shortened. Additional work is required before any more than the very rough estimates included in this report as to the limits for the several possible classifications can be made (p. 13).

In an earlier paper dated Nov. 26, 1943 [1], the author(s) had proposed a “Study to Determine the Extent to Which Individual Voice and Diction Characteristics Are Represented on the Spectrogram”. In the course of this study, recordings were to be made by five male speakers on three different dates. The recordings were to consist of (a) the numbers from 1 to 5, (b) the numbers from 6 to 0, and (3) the sentence “We shall write what you can hear”. Spectrograms were to be made of these recordings with various frequency and filter settings in the spectrograph. In a fourth recording, each speaker was to utter the sentence “We shall write what you can hear”, once in his natural voice and again in an “attempt to disguise voice and diction beyond recognition”. [...] The object of this recording is to determine if there are constants by which an individual’s voice may be identified even though he attempts to disguise it.” (Anon, p. 2) There is no evidence in the papers which this author examined to indicate that such an experiment was ever carried out. In fact, there is no record of any successful field use of spectrography as was suggested in the Gray and Kopp report [9, p. 13-14]: “It is suggested that a trial voice identification group be established to carry on work under actual or simulated field conditions to test the conclusions of this laboratory work”. There is no evidence that this kind of work was ever undertaken.

It seems that the end of WWII put a preliminary halt to the spectrograph as a tool in voice comparison. After all, the relevant documents were still classified. Consequently, all this information and knowledge had to be kept secret at the time, the sparse publications on the spectrograph maintaining the guise that it was developed to aid the deaf in using the telephone [22, 23]. It is conceivable that this procedure must have been very frustrating for the researchers.

Evidently, there was some knowledge about the speaker identification potential of the spectrograph present in the law enforcement community at the time. A memo by O.E. Buckley (FDRC) dated Jan 19, 1945 which was copied to Ralph Potter talks about an inquiry from the NYPD about voice identification [3]. He made sure to remind the recipient to keep the information about the real purpose of the spectrograph secret. The most that could be released to the NYPD “[...] was that we thought there were possibilities in this direction, but that Bell Telephone Laboratories was so heavily engaged in war work at the present time that it would be impossible to do anything until after the war.” [3, p. 1] The NYPD does not seem to have followed up on the matter after the war, unless they were the “law enforcement group” at whose request “Lawrence Kersta took up the idea of using sound spectrography for speaker identification purposes” according to Tosi [25, p. 415]. At any rate, there is no record of voiceprint identification in the early post-war years, but we have to remember that at that time, the spectrography files were still classified.

1.3 The role of Lawrence Kersta

And then came Lawrence Kersta. We do not know much about him except that he was a staff member of Bell Laboratories and that he developed an amplitude module for the spectrograph [12, 13, p.1254]. Whether or not Kersta knew the above-mentioned reports is unclear. The fact of the matter is, though, that his famous *voiceprint* article in *Nature* [13] in no way represents the knowledge and opinion of those researchers who developed the spectrograph and prepared the famous report to the US military. In essence, his claim was that all it took to carry out “voice print identification” was spectrograms of ten frequent words from both the perpetrator and a suspect, and the “expert” would then make his comparison by visually inspecting both sets. Listening to the samples was not part of the procedure (for details cf. [2]). Kersta’s paper meant a huge step backward compared to the Grey and Kopp report [9]. The same is, by the way, true for the FBI speaker identification unit of the late 20th century and its head, Bruce Koenig [16]⁴. Since we do not know whether Kersta had access to Gray and Kopp’s work [9], we cannot determine whether he was oblivious of the state of the art procedure or deliberately ignored it for his own fame and money. In view of the fact that he did not join Bell Laboratories until well after WW II and his role within that institution was not a very prominent one according to Potter et al. [22], it is fair to assume the former. Another question is why nobody stopped him. There must have been former colleagues from Bell Laboratories still around in the early 1960s who could easily have pointed out the shortcomings of the voiceprint approach, but it seems that this did not happen. Declassification of the wartime documents occurred on August 2, 1960, as is documented in the Harriet Kopp-Green files. Kersta certainly did not waste much time submitting his papers to *Nature* and JASA, which were all published in 1962 [13, 14, 15].

1.4 Interim conclusion

In summary, it can safely be stated that there was an amazingly deep understanding on the part of Gray and his group of the nature of voice comparison as well as the features which can be exploited in speaker identification. At the same time, they were well aware of the technical and situational factors affecting any speaker ID work. They also came up with the idea of collecting background data in order to assess the frequency of certain features in what could be called the relevant population, i.e. they recommended establishing a separate database for each individual language. In fact, the description of a possible methodology closely resembles what was later to become known as the auditory-acoustic method of speaker identification. None of this is preserved in the publications and – worse – in the forensic practice by Kersta and his advocates. Oskar Tosi [25], who evidently had access to the Gray and Kopp report, confines himself to citing the conclusion which is worded in a rather “catchy” way with the clear intention of catering to the military commissioners and securing more work for Bell Labs (Tosi [25, p. 415]). All words of caution and caveats which are present throughout the Gray and Kopp report are ignored. Tosi’s account thus does not do justice to the level of reflection on the part of those researchers. They were much more careful and aware of pitfalls than emerges from Tosi’s rendition. Consequently, the chapter dealing with the use of the sound spectrograph for the purpose of speaker identification during WW II needs to be rewritten.

⁴ Bruce Koenig is not to be confused with W. Koenig, who was among those developing the spectrograph.

2 Yet another set of documents

When the present author was satisfied that she had unveiled everything there was to be learned about the subtext of voiceprints, Michael Ashby made her aware of a website called *griffonage.com* and someone by the name of Patrick Feaster who actually claimed that the spectrograph was originally construed as a decryption device. Being convinced of having uncovered the ground truth about this device, she was very close to dismissing this idea as some kind of “alternative facts” created by a weirdo. Still, she took a closer look at the website, and this is what she found:

Once again, there was reference to a number of documents that were declassified in the 1960s which gave the impression of being authentic. They concern the decryption of enemy radio messages. One indication leading this author to believe in their authenticity was the person who had signed them: C.H.G. Gray of Bell Telephone Co. The only explanation for those two separate sets of documents is that there must have been two teams of scientists working independently on both aspects of the spectrograph, possibly without knowing what the other team was doing, but indirectly competing with each other. What connected both was the group leader and project supervisor C.H.G. Gray.⁵

The present author would not have been inclined to discuss this branch of research in any detail since it has little to do with phonetics at first glance had there not been a reference to spectrogram reading, which was quite popular a topic among phoneticians in the 1960s and 1970s [19, 26]. Evidently, Bell Labs tried to sell the spectrograph to the military for two different purposes in two separate projects. The second project also focuses on the ability to produce visible speech. In this case, it is suggested to use it as a device to unscramble enemy messages. And this is how it was supposed to work:

In cases where the scrambling system involves rearrangement of the speech elements in time or in both time and frequency, the basic method for determining the codes involves cutting up spectrograms along the element boundaries and rearranging the elements so as to restore the straight speech” [7, p. 41]. “The places are numbered before the matching process begins, and when the matching has been completed, the numbers on the pieces determine the code. (ibid.)

If the scrambling process involves inversion of the time or frequency scales, straight speech can be restored for matching purposes by making two spectrograms ... (ibid.)

The figures below are taken from the report by Gray and Koenig [7] and show the principle of operation (Figure 2 was post-processed for faulty scanning).

In other words, a decryption task which is characterized as extremely time critical elsewhere in the report [7, p. 57-59] requires the following steps to be taken to decrypt about 2 seconds of speech:

- 1 record the signal
- 2 rerecord it into the spectrograph
- 3 produce a spectrogram
- 4 cut up the spectrogram
- 5 rearrange the cut pieces
- 6 decode the signal by spectrogram reading

⁵ After having learned about the second project, this author found a hint at this alternative application of the spectrograph in the Gray and Kopp report: They remark in passing that “[s]crambling, such as is introduced by TDS forms of privacy, produces a spectrogram as shown in Fig. 50, but to the extent that any such scrambled message can be restored, identification can be accomplished” [9, p. 3]. (Fig. 50 is not contained in the file available to me, AB.) Also, this application of spectrography was made known to the public in the late 1970s [4].

This procedure certainly seems quite tedious. Step 3 (generating a spectrogram) alone took approximately 2.5 minutes for every 2.4 seconds of speech [6, p. 2]. Even if a “code” could be established by using this time-consuming procedure, it is promising only under the assumption that the Germans would use the same code for some time – an assumption which may not have been realistic. There is the possibility that Bell Labs tried to “sell” the spectrograph in two different contexts, just to be sure that the cost of developing it would pay off even though neither purpose seemed immediately prone to implementation. The role of the spectrograph in decryption is also briefly covered in [4, p. 293].

A note should be added on the role of the Russian scientists who developed a spectrograph of their own and used it for speaker identification purposes [2, 24]. It is now very clear that the American scientists were the first to construct such a device and propose its use for voice comparisons. The Russian scientists seem to have copied the technology including its application, but from Alexander Solschenitsyn’s account [24] it emerges that they used it on a much less sophisticated level than is described in the Gray and Kopp report of 1944 [8]. At any rate, they seem to have adhered to the “voiceprint” idea including the simplistic fingerprint analogy much more closely than the American scholars did at the time.

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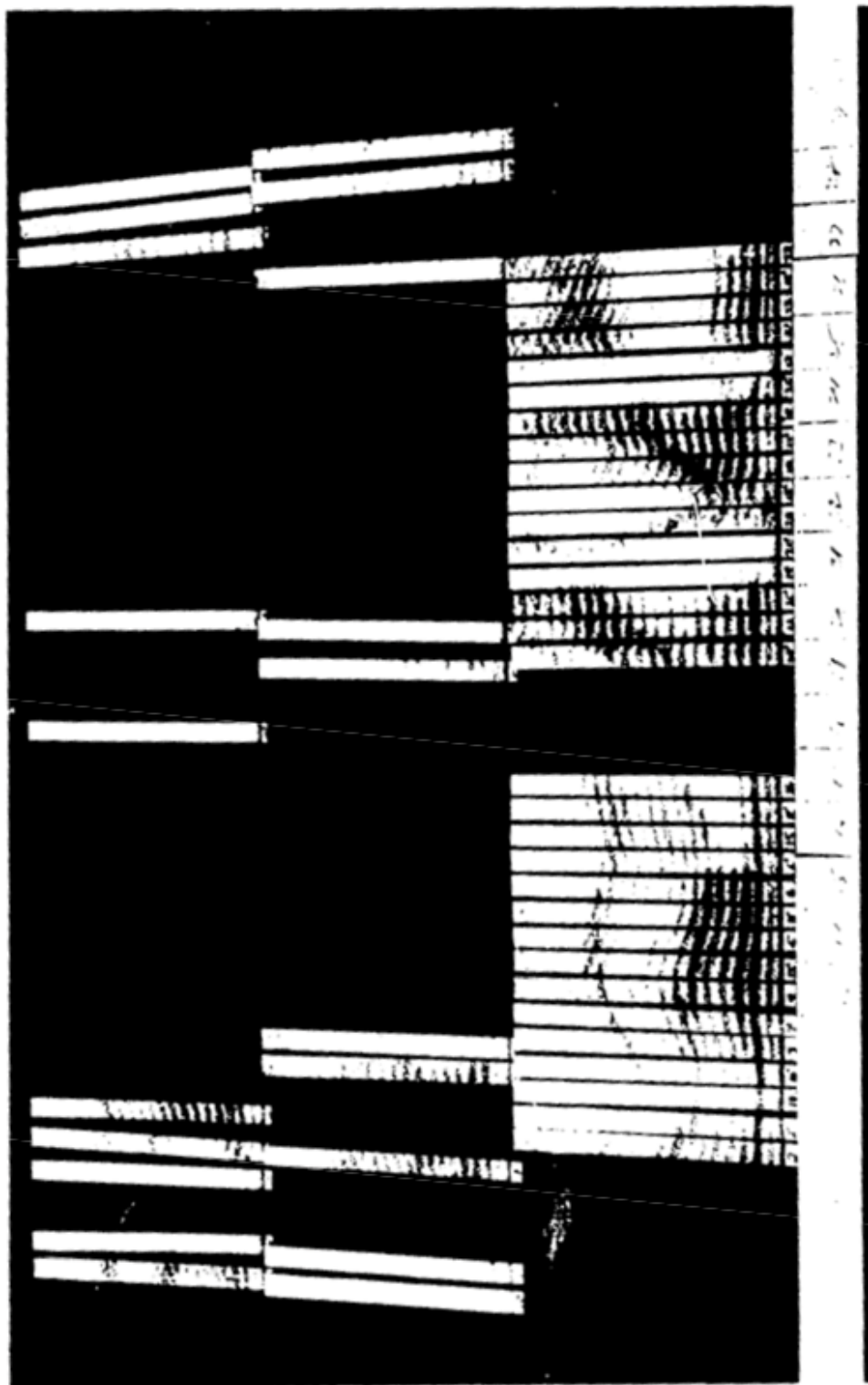
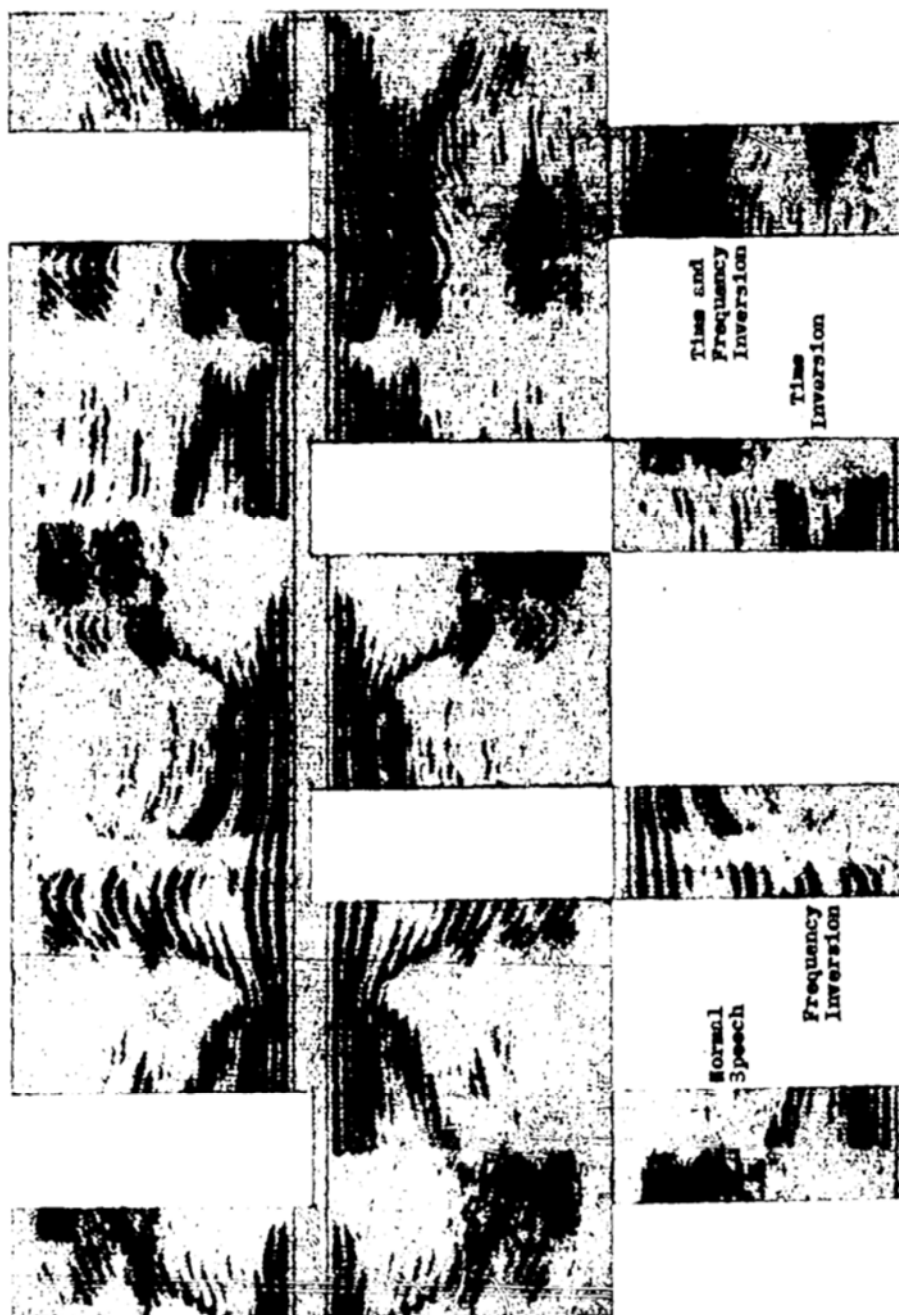


Figure 38 - Latching Spectrograph Patterns of Nonrepeated Code TDS

Figure 2. Reproduced from [6, p. 42]: Figure 38: Latching Spectrograph Patterns of Nonrepeated Code TDS.



If the scramble contains inverted elements, these will appear right side up in a mechanically inverted spectrogram.
The time scale may be inverted by rotating the elements 180 degrees. Note the positions of the base lines in the examples above.

Figure 39 - Illustrating Inversion of Time and Frequency Scales in Spectrograms

Figure 3. Reproduced from [6, p. 43]: Figure 39: Illustrating Inversion of Time and Frequency Scales in Spectrograms.

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